Air Pollution Prediction from Environmental Data

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Abstract

We compare three models for estimating hourly carbon monoxide concentration, `CO(GT)`, in the UCI *Air Quality* dataset. The data come from one urban monitoring station. Predictors include pollutant measurements, gas sensor responses, weather, hour of day, and day of week. We replaced the missing-value code -200, removed the mostly missing `NMHC(GT)` column, and filled gaps in the predictors by linear interpolation. We retained 7,674 observations with a recorded target. Ridge regression, Lasso regression, and k-nearest neighbors (KNN) were tuned by five-fold cross-validation and evaluated on a random 20% test set. KNN gave the lowest test error: RMSE $= 0.3615$ mg/m^3, MAE $= 0.2280$ mg/m^3, and $R^2 = 0.9373$. Its selected settings were five neighbors, Manhattan distance, and distance weights. Ridge and Lasso gave similar results, with R^2 close to 0.896. These results describe prediction from measurements taken at the same time; they do not establish accuracy for future hours. Interpolation before splitting, scaling before the internal validation of the linear models, and similar hours in both sets limit the evaluation. Their effects have not been measured. A chronological test is needed before using the models for future prediction. The code is available in the project repository [1].

Keywords: air quality; carbon monoxide; Ridge regression; Lasso regression; nearest neighbors; cross-validation.

1 Introduction

Air pollution affects human health, including the respiratory and cardiovascular systems [4]. Hourly monitoring data help describe how pollution changes during the day. They also allow us to test whether sensor signals and related measurements can estimate a pollutant concentration.

We study the UCI *Air Quality* dataset [2,3]. Our target is the reference carbon monoxide concentration, $Y =$ `CO(GT)`. The predictors include measurements recorded at the same station and the same hour. The task is therefore to estimate current CO from other available measurements. No future prediction horizon is defined.

We compare Ridge, Lasso, and KNN. Ridge and Lasso give simple linear models with penalties that can help when predictors are correlated. KNN predicts from nearby observations and can represent patterns that a linear model misses [7]. We assess their test errors and discuss what can be learned from this comparison. The study concerns prediction, so the fitted relations should not be read as causal effects.

2 Dataset and Exploratory Analysis

2.1 Dataset structure

The source file contains 9,358 dated hourly records from an urban station in Italy, from 10 March 2004 to 4 April 2005. It also contains 113 empty rows, which do not enter the analysis. It combines reference pollutant concentrations, responses from five gas sensors, and weather measurements [2]. Carbon monoxide is measured in mg/m^3.

After excluding the target and the mostly missing `NMHC(GT)` column, the predictors are:

- reference concentrations: `C6H6(GT)`, `NOx(GT)`, and `NO2(GT)`;
- sensor responses: `PT08.S1(CO)`, `PT08.S2(NMHC)`, `PT08.S3(NOx)`, `PT08.S4(NO2)`, and `PT08.S5(O3)`;
- temperature `T`, relative humidity `RH`, and absolute humidity `AH`;
- hour of day (0–23) and day of week (Monday $= 0$ to Sunday $= 6$).

This gives 13 predictors. The full date and timestamp are excluded. Hour and day of week are treated as numerical predictors in the code. This does not represent their cyclic nature: for example, hours 23 and 0 are close in time but far apart numerically. Because the models use other reference pollutant measurements, this task is broader than estimating CO from low-cost sensors alone.

2.2 Missing values and extreme observations

Missing measurements are coded as -200. We replaced this code with missing values before any analysis. We removed `NMHC(GT)` because most of its observations were missing. Rows without an observed target were excluded from model fitting and evaluation.

An interquartile-range rule flagged several extreme values. The workflow retained these observations. High concentrations may be real pollution events. Removing them automatically would change the task and could hide errors during the hours that matter most.

2.3 Daily and weekly patterns

The hourly distributions suggest two broad pollution peaks, one around 8–9 a.m. and another between 6 p.m. and 8 p.m. These times are consistent with morning and evening traffic. However, the data do not include traffic counts, so the profiles alone cannot identify their cause.

Temperature tends to peak later in the day, while relative humidity follows a different pattern. The average profiles also suggest lower pollution at weekends than on weekdays. These are descriptive patterns. Weather, season, and other differences between the observed days

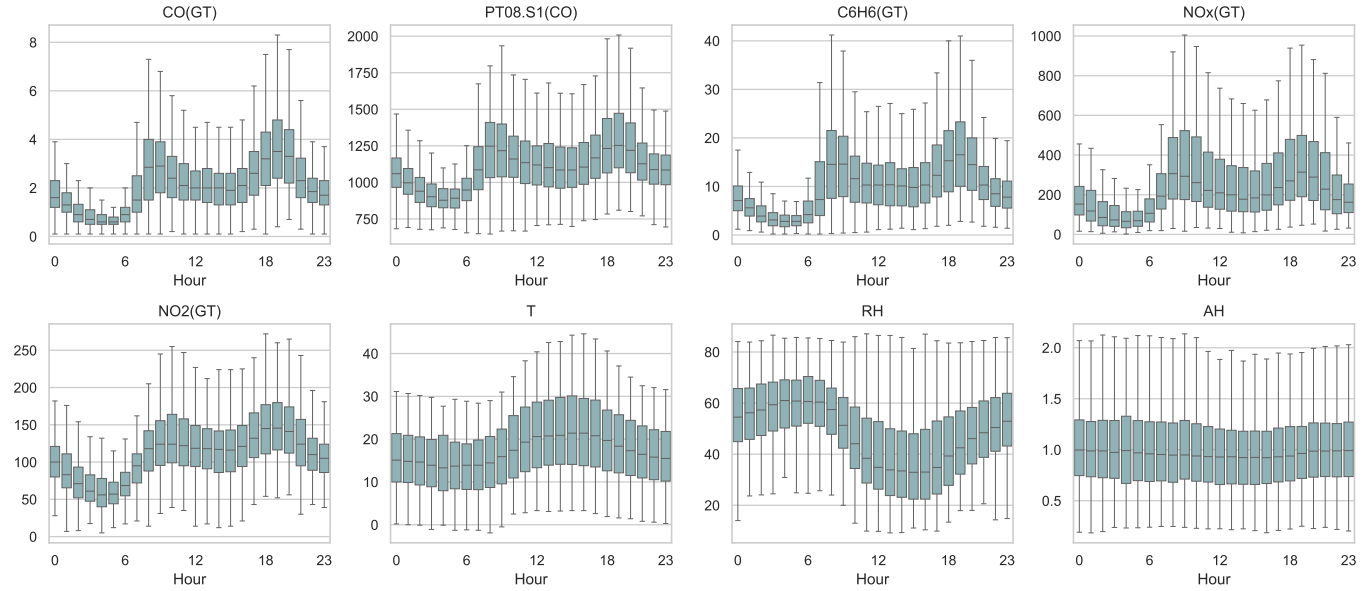


Figure 1: Hourly distributions. CO is in mg/m^3; benzene and NO_2 in μg/m^3; NO_x in ppb; temperature in °C; and relative humidity in percent. Sensor response and absolute humidity use the dataset scales. The pollution peaks are consistent with rush-hour activity, but do not establish its effect.

may also contribute.

2.4 Correlations

The largest positive Pearson correlations with `CO(GT)` are approximately 0.910 for `C6H6(GT)`, 0.895 for `PT08.S2(NMHC)`, and 0.863 for `PT08.S1(CO)`. The correlation with `PT08.S3(NOx)` is about -0.696. The weather variables have much smaller marginal linear correlations, with absolute values below about 0.06.

These correlations describe pairs of variables after cleaning. They do not measure a variable's contribution once the other predictors are included. A small Pearson correlation can also miss a nonlinear relation or an effect that appears after a delay.

Several predictors are strongly correlated with each other. This can make ordinary least-squares coefficients sensitive to changes in the sample. Ridge and Lasso address this problem through regularization, although neither method makes correlated coefficients automatically easy to interpret [5,6].

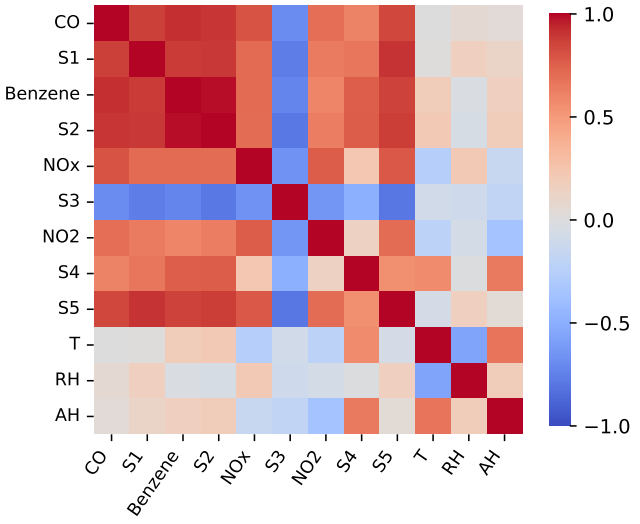


Figure 2: Pearson correlations after cleaning. S1–S5 denote the PT08 sensor responses. Pairwise correlations do not establish variable importance or causality.

3 Data Preprocessing

3.1 Cleaning and interpolation

After replacing -200, removing `NMHC(GT)`, and excluding rows with an unknown target, 7,674 observations remained. We filled gaps in the remaining predictors by linear interpolation along the record order.

For two observed values at row positions $r_0 < r_1$, interpolation estimates a missing value at position r by

$$x_r = x_{r_0} + \frac{r - r_0}{r_1 - r_0}(x_{r_1} - x_{r_0}).$$

The code also fills leading or trailing gaps from the nearest available value. This is a simple way to fill short gaps, but it can smooth sharp peaks and give poor estimates across long gaps. Rows without CO have already been removed, so adjacent rows may be more than one hour apart. Interpolation by row position is therefore not the same as interpolation by elapsed time.

3.2 Training, validation, and testing

We randomly assigned 80% of the cleaned observations to training and 20% to testing, with random seed 42. This gave 6,139 training observations and 1,535 test observations. Five-fold cross-validation on the training set selected the model settings. The folds were shuffled with the same seed. For each candidate, the model was fit-

ted on four folds and evaluated on the remaining fold. The validation results were then averaged. The selected model was fitted again on the training set and evaluated on the test set.

The Ridge search used 100 logarithmically spaced penalty values from 10^{-3} to 10^{3}. The Lasso search used 100 values from 10^{-3} to 10^{1}. KNN tested $k = 2, \ldots, 30$, uniform or distance weights, and Manhattan or Euclidean distance. The selection criteria differed: mean validation R^2 for Ridge, mean squared error for Lasso, and mean RMSE for KNN.

The code handles scaling differently across models. KNN is fitted through a grid search around the full pipeline, so its scaler is fitted separately within each fold. For Ridge and Lasso, the scaler is fitted once on all training observations before the model's internal cross-validation. Their validation folds therefore contribute to the scaling parameters, which is a source of leakage within the training set. The test observations do not enter the scaler fit. A stricter version would place the whole Ridge or Lasso pipeline inside the cross-validation search as well.

Interpolation was performed before the split. A filled training value may therefore depend on a value later assigned to validation or testing. This is a source of information leakage, and its effect on the reported scores is unknown. The random split creates a second concern: adjacent hours can fall into different sets even when their measurements are very similar. This matters especially for a model such as KNN, which relies on nearby observations.

The scores should therefore be read as results from a random split of an interpolated series. A stricter evaluation would use an earlier period for training, a later period for testing, and an imputation rule that uses only information available at prediction time. A gap between periods could also reduce dependence between the training and test records.

3.3 Standardization

For predictor j, standardization gives

$$z_{ij} = \frac{x_{ij} - \mu_{j,\text{train}}}{s_{j,\text{train}}},$$

where the mean and standard deviation are estimated from the data used to fit the scaler. As described above, these are the fold's training observations for KNN and the full training set for Ridge and Lasso.

This puts the predictors on comparable scales. In Ridge and Lasso, it prevents measurement units from determining the relative strength of the penalty. In KNN, it prevents a variable with a large numerical range from dominating the distance.

4 Modeling Methods

Ridge and Lasso fit one linear relation across the sample. KNN uses a different local neighborhood for each prediction. The models were implemented with scikit-learn [8]. In the following equations, n denotes the number of observations used for fitting, z_i is a standardized predictor vector, and b is an unpenalized intercept.

4.1 Ridge regression

Ridge minimizes

$$\min_{b,\beta} \left\{ \sum_{i=1}^{n} (y_i - b - z_i^\top \beta)^2 + \alpha \sum_{j=1}^{p} \beta_j^2 \right\}.$$

The penalty shrinks the coefficients towards zero. This can reduce the variation caused by correlated predictors, at the cost of some bias [5].

Cross-validation selected $\alpha \simeq 1.8738$. Ridge gave a test RMSE of 0.4652 mg/m^3 and $R^2 = 0.8961$. It therefore captured much of the variation in this test set using a global linear model.

4.2 Lasso regression

Lasso uses an absolute-value penalty, which can set coefficients exactly to zero [6]. With the scikit-learn normalization, it minimizes

$$\min_{b,\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - b - z_i^\top \beta)^2 + \alpha \sum_{j=1}^{p} |\beta_j| \right\}.$$

The different loss normalizations mean that Ridge and Lasso values of α should not be compared directly.

The selected value was $\alpha = 0.001$, the lower end of the search grid. All fitted coefficients remained nonzero, so Lasso did not produce a smaller predictor set in this run. Its test RMSE was 0.4657 mg/m^3, close to the Ridge result. Testing smaller penalties would show whether the search boundary affected this choice.

When predictors are strongly correlated, several coefficient combinations may give similar predictions. Lasso coefficients can then change across samples. A nonzero coefficient does not establish that a variable has a separate physical effect.

4.3 k-nearest neighbors

KNN predicts from the responses of nearby training observations [7]. With uniform weights,

$$\hat{y}(z) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(z)} y_i,$$

where $\mathcal{N}_k(z)$ is the set of k nearest neighbors. With distance weights,

$$\hat{y}(z) = \frac{\sum_{i \in \mathcal{N}_k(z)} w_i(z) y_i}{\sum_{i \in \mathcal{N}_k(z)} w_i(z)}.$$

For positive distances, $w_i(z) = 1/d(z, z_i)$. Exact matches are handled separately so that division by zero is avoided.

The selected model had a mean five-fold validation RMSE of 0.3898 mg/m^3. It used $k = 5$, distance weights, and Manhattan distance:

$$d(z, u) = \sum_{j=1}^{p} |z_j - u_j|.$$

Table 1: Test scores from the random split. RMSE and MAE are in mg/m^3. KNN uses five neighbors, Manhattan distance, and distance weights.

Model	RMSE	MAE	R^2
Ridge	0.4652	0.2961	0.8961
Lasso	0.4657	0.2955	0.8959
KNN	0.3615	0.2280	0.9373

This corresponds to the Minkowski parameter $p = 1$ in the implementation; this distance parameter is separate from the number of predictors used in the formulas above.

KNN gave the lowest test error of the three models. This is consistent with useful local structure in the data. However, it may also benefit from the temporal similarity between training and test observations. The present comparison cannot separate these explanations.

5 Model Evaluation

For m test observations, we calculated

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \widehat{y}_i)^2},$$

$$\text{MAE} = \frac{1}{m}\sum_{i=1}^{m}|y_i - \widehat{y}_i|,$$

and

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{m}(y_i - \bar{y}_{\text{test}})^2}.$$

RMSE gives more weight to large errors than MAE. Both are expressed in the target unit, mg/m^3. The R^2 score compares squared prediction error with variation around the test-set mean.

KNN had the lowest RMSE and MAE and the highest R^2 (Table 1). Its RMSE was about 22% lower than Ridge's. Ridge and Lasso had nearly equal errors. Their similarity suggests that the choice between these two penalties made little difference to prediction in this run.

These are results from one test split. We did not calculate uncertainty intervals for the differences between models, so the table does not show how much their scores would change under another split or in another period.

5.1 Effect of model settings

For Ridge and Lasso, a stronger penalty gives more shrinkage. Too little shrinkage can leave coefficients sensitive to correlated predictors; too much can remove useful signal. Cross-validation chooses among the values included in the search.

For KNN, a small k uses a narrow neighborhood and can follow local variation closely. A larger k gives smoother predictions but may miss local patterns. Distance weights can make one or two very close observations dominate the estimate. The selected value of five neighbors is therefore a local fit, and its behavior should be checked in later periods.

6 Interpretation and Discussion

6.1 Which measurements carry information?

The strongest pairwise links with CO involve benzene and several sensor responses. They carry information about the pollutant mixture measured at the same station. Their high correlations do not provide a ranking of independent contributions because the predictors also overlap in the information they carry.

Temperature and humidity have weak marginal linear correlations with CO in this sample. This does not show that weather is unimportant. Its relation with pollution may depend on other variables, appear after a delay, or take a nonlinear form. A separate analysis would be needed to test these possibilities.

6.2 Interpreting the models

The linear models offer a compact description. With standardized predictors and an unscaled target, a coefficient is the fitted change in CO associated with a one-standard-deviation increase in that predictor, holding the other predictors fixed. Strong correlations can make this interpretation fragile, and shrinkage also affects coefficient size. Coefficients should therefore be read as parts of a fitted equation rather than direct measures of physical importance.

KNN has no single coefficient for each variable. Its predictions can instead be examined through the selected neighbors, their distances, and their recorded CO values. This is useful for individual cases but does not give one simple relation for the whole dataset.

6.3 Prediction and causality

The models aim to predict CO from the observed predictors. They do not estimate what would happen if one pollutant or weather variable were changed by an intervention. For example, a strong benzene–CO association may reflect shared emission sources. It does not show that benzene causes CO.

6.4 Limits and next steps

The main limitation is the evaluation design. Interpolation before splitting allows information to cross sample boundaries, and a random split places related hours in both sets. Ridge and Lasso also use validation-fold information for scaling. We cannot assume that these effects are small. A chronological evaluation with an appropriate imputation rule and scaling fitted within every fold is the most useful next step. A common model-selection score would also make the tuning procedures easier to compare.

All observations also come from one station. Scores may change in a later season, at another location, or after changes in the sensors. The use of reference pollutant concentrations makes the current task different from a system that has only low-cost sensors and weather data. Those reference measurements would need to be available when a prediction is made.

Further comparisons could therefore test two predictor

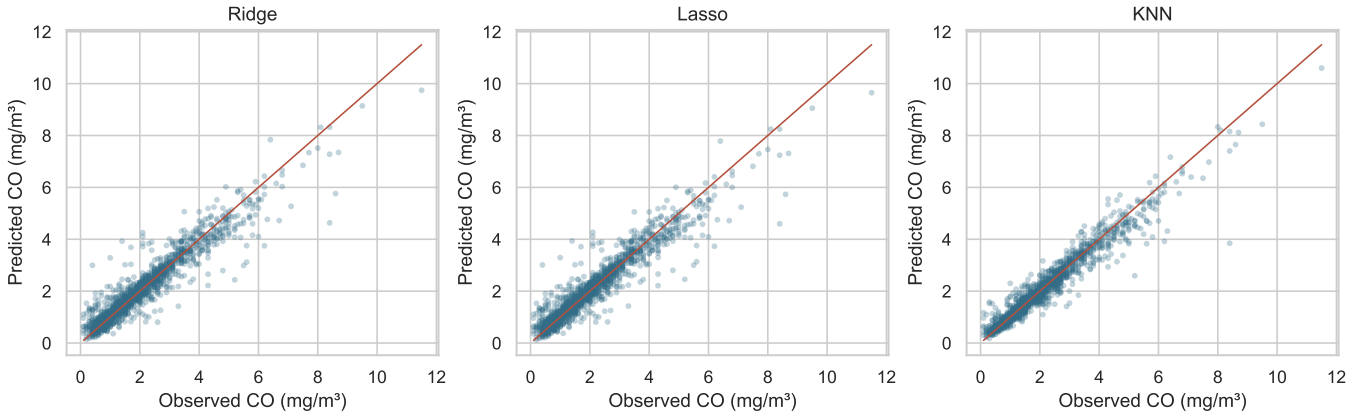


Figure 3: Observed and predicted CO concentrations in the test set. Points on the diagonal represent exact predictions. This plot describes agreement in the selected test sample.

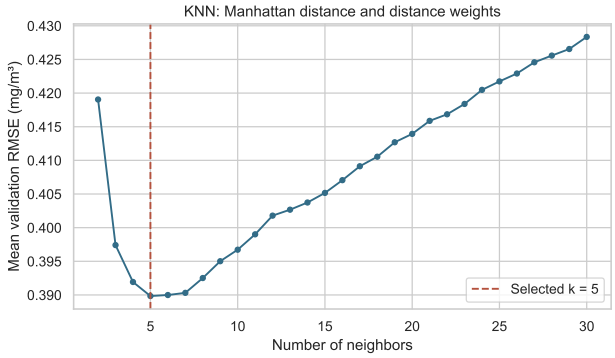


Figure 4: Cross-validated KNN RMSE for Manhattan distance and distance weights. The vertical line marks the selected neighborhood size.

sets: all current measurements, and sensors plus weather alone. Repeated temporal evaluations, errors during high-CO events, and tests at another station would give a clearer picture of practical performance.

7 Conclusion

The measurements in this dataset allow accurate estimates of CO within the reported random test split. KNN performed best, with RMSE 0.3615 mg/m^3, MAE 0.2280 mg/m^3, and $R^2 = 0.9373$. Ridge and Lasso gave similar, less accurate predictions and provided simpler global equations. Lasso kept all predictors at the selected penalty.

These findings concern estimates from measurements taken at the same hour at one station. They do not yet establish performance for future hours or other locations. Interpolation before splitting, scaling before internal validation for Ridge and Lasso, and temporal dependence limit the evaluation. A chronological test with fully separated preprocessing is needed to determine whether KNN keeps its advantage when the model is applied to a later period.

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